$^{40}Ar/^{39}Ar$ dating of a Late Proterozoic palaeomagnetic pole for the Armorican Massif (France)

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SUMMARY

As part of a combined geochronologic and palaeomagnetic study of the Trégor area (Armorican Massif), we have attempted to obtain a reliable age for a Late Proterozoic palaeopole from experiments carried out on the Keralain and Petit Taurel diorites.

A heterogeneous population of amphiboles analysed by the $^{40}Ar/^{39}Ar$ step-heating method gave a flat saddle-shaped age spectrum (characteristic of excess argon) with a minimum age of 604.9 ± 1.2 Ma, which represents a maximum estimate of the age of the mineral. The isochron calculation gives an initial $^{40}Ar/^{36}Ar$ ratio sensibly higher than the atmospheric ratio. This indicates that the isochron age, 583 ± 2 Ma, is a better estimate of the age of the Keralain diorite. Laser single step fusion and step-heating experiments performed on single grains representative of the different components of this heterogeneous population of amphiboles show that the excess argon affects it in its entirety. The minimum age of 590.8 ± 2.3 Ma of the flat saddle-shaped age spectrum displayed by the core of a crystal of hornblende is concordant (at the 2σ level) with the isochron age. The isotopic closure temperature calculated from the step-heating results is 556 ± 43 °C.

The palaeomagnetic study has shown, using a thermal demagnetization procedure, that the samples have a multicomponent magnetization. Two characteristic components (A and B) have been recognized respectively at Keralain and the Petit Taurel. A careful rock magnetic study has established that the magnetic carrier of the primary component A, isolated at Keralain, is magnetite with a magnetic blocking temperature of 575 °C. The similarity of this temperature with the calculated isotopic closure temperature allows us to assign an age of *c.* 583 Ma to the corresponding well-defined palaeopole 'DKA', located at 33°N, 256°E. The demagnetization experiments have also clearly shown that component B, well isolated at the Petit Taurel, is a remagnetization probably related to magmatic events at around 550 Ma. Furthermore, its magnetic behaviour, as observed at Keralain, indicates a possible thermochemical origin.

The well-dated DKA palaeopole allows us to refine the late Proterozoic polar wander path for Armorica and seems to confirm that the Armorican Massif and southern United Kingdom were linked together at that period.

Key words: excess argon, isotopic and magnetic closure temperatures, laserprobe dating, palaeomagnetism, Proterozoic.

INTRODUCTION

During the last twenty years, numerous geochronologic studies have allowed us to clarify the succession of magmatic events that characterize the Cadomian orogeny in the Armorican Massif (western France). It appears (Graviou *et al.* 1988) that a first period of calc-alkaline magmatic activity (between 660 and 580 Ma) was restricted to the Domnonean domain (Cogné 1974). Several palaeomagnetic studies of rocks related to this event have been published but none of them corresponds to units precisely dated.

The present work is a combined geochronologic

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($^{40}Ar/^{39}Ar$ step-heating technique) and palaeomagnetic study of the Keralain diorite, with the aim of obtaining a reliable age for a Late Proterozoic pole for the Armorican Massif, through a comparative study of magnetic and isotopic blocking temperature (Berger & York 1979; Berger & York 1981a; Baksi 1982).

While a palaeomagnetic study has already been performed on the Keralain diorite by Hagstrum *et al.* (1979), we have entirely resumed that study and extended it to the similar diorite of the Petit Taurel, in order to constrain the structural control, which is generally difficult for a plutonic massif.

In addition to the classical $^{40}Ar/^{39}Ar$ step-heating technique, laser-probe analysis using a continuous argon-ion laser, a new microdating methodology pioneered by York *et al.* (1981) and Layer, Hall & York (1987), was used to locate excess argon in the heterogeneous population of amphiboles used to date the Keralain diorite.

GEOLOGY AND SAMPLING

This work concerns two dioritic stocks, each with a maximum diameter of 100 m, located in the Trégor, northernmost part of Brittany (Fig. 1), which belongs to the Domnonean domain of the Armorican Massif (Cogné 1974). One is near Keralain (1), the other forms the islet of Petit Taurel at the headland of Plouézec (2). At Keralain, the stock seems intrusive into the lower Brioverian volcano sedimentary series without any visible contact. At the Petit Taurel, dykes emanating from the stock and intersecting the Brioverian sediments, can be observed, together with a contact metamorphism in the sediments at the diorite contact.

In the early Palaeozoic times (the Eocambrian), an acid and alkaline magmatic episode occurred, represented by various forms. These are the granites of Porz-Scarff and Paon as well as the microgranitic dykes of Loguivy in the North Trégor basement, and the ignimbritic effusion of Lézardrieux, which overlies the Brioverian sediments in the South Trégor domain. These ignimbrites, limited today to a few outcrops, probably originally had a much greater extent (Lefort 1970). The North and South Trégor structural framework was produced during the Hercynian tectonogenesis, with the development of two major faults trending E–W.

The stocks of Keralain and Petit Taurel have approximately the same mineralogical composition. The main constituents are a calcic hornblende (60 per cent) and a calcic plagioclase (20 per cent). These bodies are considered as being the result of a fractional crystallization of dioritic magmas at the origin of the more important dioritic intrusions, located southwards (diorite de Saint Brieux) (Auvray 1979). Together with the calc-alkaline suite of the North Trégor Batholith, these instrusions characterize the geotectonic context of the Brioverian (Graviou *et al.* 1988; Guerrot & Peucat 1990).

Because of the small size of the two dioritic stocks, only one sampling site was possible in each body: 31 palaeomagnetic samples were taken at Keralain (Site 1) and nine at the islet of the Petit Taurel (Site 2) (Fig. 1). The samples were drilled in the field and oriented by magnetic and sun

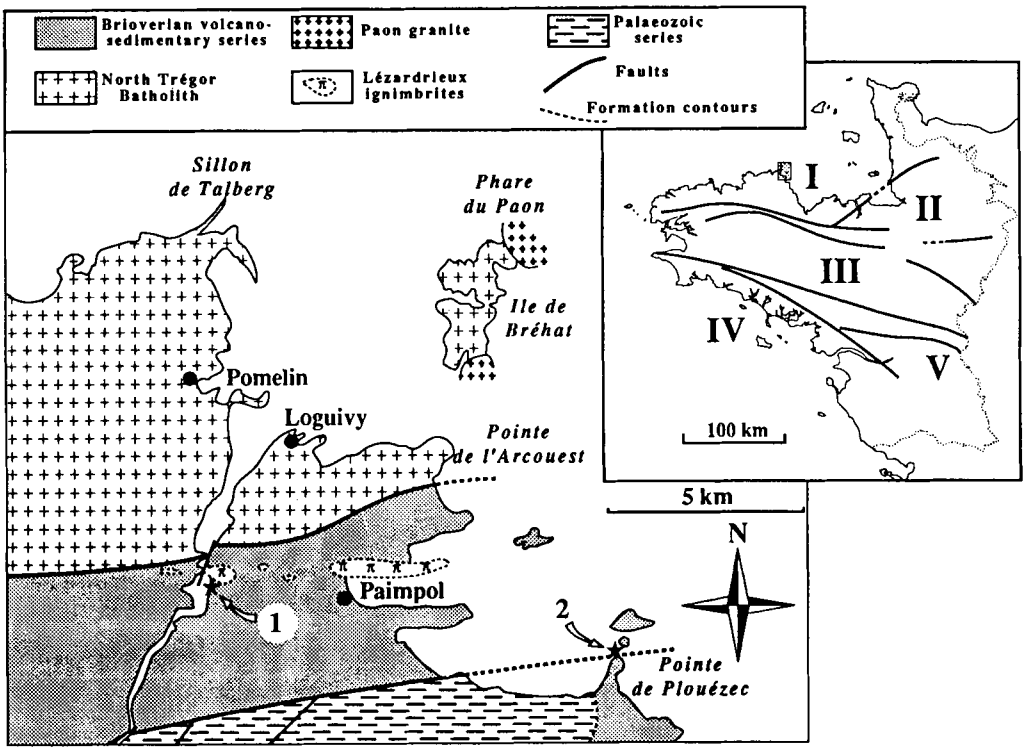


Figure 1. Sampling localities and schematic geological features of the Tregor area (Auvray 1979), in the Domnonean domain (I) (Cogné 1974), Brittany (France). Inset: (I) Domnonean domain, (II) Mancellian domain, (III) Centro-Armorican domain, (IV) South Armorican domain, (V) Ligerian domain.

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compasses, (the latter, weather permitting). The palaeomagnetic samples for each site were spaced out all along the stocks. One hand specimen of the Keralain diorite (1) was sampled for geochronology approximately at the centre of the stock.

GEOCHRONOLOGY

Sample description

The amphiboles from the Keralain diorite are the only suitable minerals for $^{40}Ar/^{39}Ar$ geochronology. These are dominated by hornblende that often presents an optic zonation with green colours at the rim and brown at the core. An electron microprobe traverse across a zoned hornblende has shown that from the core to the rim of this mineral, the Ca/K ratio increases from 5 to 11 with a composition ranging from that of a ferroan pargasite to an edenitic hornblende (Leake's classification; Leake 1968). Further electron microprobe analysis revealed the existence of green actinolite or actinolitic hornblende characterized by higher Ca/K ratios.

Experimental procedure

For stepwise heating experiments, the 160–210 μm fraction of amphiboles was separated by standard methods (magnetic separator and heavy liquids). The final mineral separate, named 87CAS, was obtained by hand-picking and contained mostly brown amphiboles but also some green amphiboles. For laser experiments, brown and green pieces of amphiboles were selected from the 0.5–2 mm fraction range. The samples were irradiated in the Mélusine reactor (C.E.N. Grenoble) (Féraud *et al.* 1986), with the standard LP-6 biotite (128.5 Ma; Roddick 1983) and in the Osiris reactor (C.E.N. Saclay) with the standard MMhbl (520.4 Ma; Samson & Alexander 1987). In the Osiris reactor, the samples were irradiated with a total flux of 9×10^{18} neutrons cm^{-2} (with about 10 per cent of fast neutrons) and were rotated during irradiation. The standard $^{40}Ar/^{39}Ar$ step-heating analysis (described by Féraud *et al.* 1982) of the sample 87CAS was performed with a radio-frequency fusion system. The temperature error is estimated at ±20 °C below 900 °C and ±15 °C above 900 °C. The duration of each heating step was 45 ± 3 min. For the laser step-heating experiments, the laser beam size must be at least 2.5 times greater (up to 6 mm) than the sample size to obtain an homogeneous temperature on the whole grain. The $^{40}Ar/^{39}Ar$ laser microprobe analysis procedure is described by Scaillet *et al.* (1990) and Ruffet & Féraud (1990). The mass spectrometer system includes a 120° M.A.S.S.E. tube, a Baür-signer source and either (1) a Faraday cup collector [for the classical step heating experiment (87CAS) and the laser single-step fusions (87CAL)] or (2) an E.T.P. 1000 electron multiplier (for the laser step-heating experiments 89CAN and 89CAV). The results are summarized in Table 1. Errors are quoted at the 1σ level and do not include the errors on the $^{40}Ar^*/^{39}Ar_K$ ratio and age of the monitor.

RESULTS

Classical radio-frequency step-heating experiment (87CAS)

The age spectrum shows discrepant ages between 600 °C and 900 °C and a regular saddle shape from 950 °C up to 1600 °C with a minimum age of 604.9 ± 1.2 Ma (see Fig. 2). The $^{37}Ar_{Ca}/^{39}Ar_K$ ratio, proportional to the Ca/K ratio ($^{37}Ar_{Ca}/^{39}Ar_K = 0.56 \times Ca/K$) (Féraud *et al.* 1982) also shows very variable values between 600° and 900 °C. Above 950 °C, there is a regular decrease from high values of the $^{37}Ar_{Ca}/^{39}Ar_K$ ratio (Ca/K = 26.1) until reaching a plateau (Ca/K ≈ 6.8). The low-temperature (600–900 °C) ages probably correspond to K-rich secondary products whereas the saddle-shaped part of the age spectrum is given by the amphiboles themselves. The saddle-shaped age spectra are well known to be characteristic of excess ^{40}Ar (Harrison & McDougall 1981) and the minimum age is generally interpreted as a maximum estimate of the true age (Lanphere & Dalrymple 1976).

The correlation diagram ($^{36}Ar/^{40}Ar$ versus $^{39}Ar/^{40}Ar$) (Turner 1971; Roddick, Cliff & Rex 1980; Hanes, York & Hall 1985), which is an alternate form of the classical isochron analysis (Merrihue & Turner 1966) has been calculated on the eight steps from experiment 87CAS between 1000 °C and 1280 °C (Fig. 3). The low-temperature steps, corresponding to secondary phases, were rejected, as well as the high-temperature steps, because there was an important contribution from the blank of the extraction system. Using the York's fitting method (York 1969), we derived an initial $^{40}Ar/^{36}Ar$ ratio of 862(+43/−39). The corresponding calculated age is 583 ± 2 Ma. Both the age, sensibly lower than the minimum age of the saddle-shaped age spectrum, and the high initial $^{40}Ar/^{36}Ar$ ratio, confirm the excess argon already suggested by the shape of the age spectrum.

The statistical parameter MSWD (mean square of weighted deviates) (lower than 1) indicates the goodness of fit of the line and suggests, as proposed by Roddick (1978), no geological disturbance after the creation of the stock.

These results are confirmed by those obtained from the classical isochron calculation (Fig. 3).

Laser-probe analysis

Three laser experiments were performed on amphiboles from the diorite of Keralain.

Laser single-step fusions of fragments of amphiboles

Total laser fusions were performed on pieces of green (87CAL2: F–J) and brown (87CAL1: A–E) amphibole. All the experiments are fusions of one grain except for experiments 87CAL2H and 87CAL2I for which they were respectively two and four grains.

The results (Fig. 4) show that (1) the pieces of brown amphibole, characterized by low and homogeneous Ca/K ratios (≈5.7), give ages from 590 ± 12 Ma to 630 ± 19 Ma (weighted mean* of 609 ± 5 Ma) and (2) the pieces of green amphibole, in spite of some larger error bars due to a very

* Arithmetical mean of the laser single step fusion ages weighted by the inverse of their respective variances.

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Table 1. $^{40}Ar/^{39}Ar$ analytical data for (1) incremental heating experiment on sample 87CAS, (2) laser single-step fusion experiments on brown amphiboles [87CAL1 (A–E)] and green amphiboles [87CAL2 (F–K)]. (3) laser step-heating experiment on a brown fragment of amphibole and (4) laser step-heating experiment on a green fragment of amphibole. ^{37}Ar is corrected for decay, and uncertainties are quoted at the level of one standard error. The ratios used to correct interfering isotopes are listed in Féraud *et al.* (1982). Isotopic constants used are those suggested by Steiger & Jäger (1977).

Temperature (°C)	^{39}Ar %	$^{37}Ar_{Ca}/^{39}Ar_K$	Atmospheric contamination %	$^{40}Ar^*/^{39}Ar_K$	Apparent age (Ma)
87CAS	**Amphiboles** (population)				
600	1.03	2.94	49.96	38.76 ± 0.25	969.3 ± 4.9
750	1.77	1.11	45.53	31.44 ± 0.17	821.8 ± 3.6
900	3.56	2.22	44.84	20.89 ± 0.08	585.5 ± 2.1
950	1.19	14.62	22.27	26.74 ± 0.16	720.3 ± 3.6
1000	3.03	9.55	9.47	25.91 ± 0.10	701.7 ± 2.5
1040	6.97	5.59	4.19	22.55 ± 0.06	624.6 ± 1.6
1080	16.18	4.13	2.35	21.79 ± 0.05	606.9 ± 1.3
1120	27.06	3.72	2.27	21.71 ± 0.04	604.9 ± 1.2
1160	10.87	3.84	2.60	21.95 ± 0.05	610.7 ± 1.4
1200	17.40	3.72	2.77	22.00 ± 0.05	611.8 ± 1.4
1240	3.03	3.78	4.26	22.10 ± 0.08	614.2 ± 1.9
1280	6.91	3.79	3.28	22.07 ± 0.04	613.5 ± 1.2
1380	0.83	3.76	58.87	22.71 ± 0.25	628.5 ± 5.9
1600	0.17	3.83	77.74	26.03 ± 1.17	705. ± 26.
				Total age = 620.9 ± 0.5	
89CAN	**Brown amphibole** (single grain)				
1	0.51	2.71	21.16	26.30 ± 1.63	709. ± 36.
2	0.57	1.12	14.52	19.17 ± 1.54	543. ± 38.
3	0.97	2.14	8.10	19.82 ± 0.43	558. ± 10.
4	1.18	0.95	9.59	17.19 ± 0.45	493. ± 11.
5	1.55	0.57	4.28	16.93 ± 0.26	486.7 ± 6.6
6	2.16	0.37	3.32	17.57 ± 0.20	502.8 ± 5.0
7	3.76	0.47	2.29	18.61 ± 0.16	528.7 ± 4.0
8	5.52	0.60	0.97	19.79 ± 0.12	557.3 ± 2.9
9	4.57	0.85	1.38	20.04 ± 0.11	563.6 ± 2.8
10	2.56	1.92	1.88	20.23 ± 0.22	568.0 ± 5.3
11	1.52	1.90	0.50	20.46 ± 0.33	573.6 ± 7.9
12	2.91	2.55	0.36	21.01 ± 0.16	586.8 ± 3.7
13	5.83	2.97	1.15	21.34 ± 0.13	594.7 ± 3.1
14	15.18	3.19	1.07	21.92 ± 0.06	608.2 ± 1.4
15	17.34	3.16	0.60	21.68 ± 0.07	602.6 ± 1.8
16	11.89	3.12	0.51	21.44 ± 0.07	596.9 ± 1.8
17	4.68	3.09	0.99	21.18 ± 0.10	590.8 ± 2.3
18	0.53	3.01	4.23	20.41 ± 0.91	572. ± 22.
19	0.35	3.18	7.36	19.55 ± 1.67	552. ± 41.
20	16.42	3.22	0.94	21.51 ± 0.07	598.5 ± 1.7
				Total age = 586.4 ± 0.8	
89CAV	**Green amphibole** (single grain)				
1	16.66	0.69	15.32	13.98 ± 0.04	410.8 ± 1.2
2	8.42	1.88	8.04	14.91 ± 0.05	435.0 ± 1.2
3	4.65	6.54	18.39	16.13 ± 0.11	466.6 ± 2.7
4	4.88	17.21	27.03	18.59 ± 0.11	528.2 ± 2.8
5	5.82	19.81	19.92	19.68 ± 0.16	554.8 ± 3.8
6	6.38	15.29	12.35	19.10 ± 0.09	540.6 ± 2.2
7	6.95	10.51	7.04	18.95 ± 0.08	537.0 ± 1.9
8	3.69	11.05	7.59	18.20 ± 0.13	518.5 ± 3.1
9	42.57	9.49	6.52	21.06 ± 0.05	588.0 ± 1.4
				Total age = 526.0 ± 0.7	

low amount of argon released, give very variable but higher ages (up to 1.2 Ga) that seem directly correlated with the Ca/K measured ratios that range from 20 up to 90.

Laser step heating of a fragment of brown amphibole (89CAN; see Fig. 2.)

The integrated age† is 586.4 ± 0.8 Ma. After a very rapid decrease of the ages from 710 to 490 Ma in the first five steps (4.8 per cent of the total $^{39}Ar_K$ released), the age spectrum shows a regular increase of the ages (steps 6–14) up to 608.2 ± 1.4 Ma followed by a regular and flat saddle-shape with a minimum age of 590.8 ± 2.3 Ma. The corresponding $^{37}Ar_{Ca}/^{39}Ar_K$ spectrum is directly correlated to the age spectrum with a constant value ($Ca/K \approx 5.6$) for the last 66 per cent of $^{39}Ar_K$ released. This flat part of the $^{37}Ar_{Ca}/^{39}Ar_K$ spectrum, corresponding to the saddle shape of the age spectrum, seems to characterize the brown amphibole.

† Arithmetical mean of the apparent ages of each step weighted by the corresponding relative fraction of $^{39}Ar_K$ relased.

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Table 1. (*continued*)

Temperature (°C)	^{39}Ar %	$^{37}Ar_{Ca}/^{39}Ar_K$	Atmospheric contamination %	$^{40}Ar^*/^{39}Ar_K$	Apparent age (Ma)
87CAL1-A	**Brown amphiboles**	(total fusions)			
Fuse	100	3.19	4.72	22.12 ± 0.56	614. ± 13.
87CAL1-B	"				
Fuse	100	3.00	0.66	22.83 ± 0.83	630. ± 19.
87CAL1-C	"				
Fuse	100	2.76	1.42	21.62 ± 0.32	601.7 ± 7.7
87CAL1-D	"				
Fuse	100	3.31	5.37	22.77 ± 0.48	629. ± 11.
87CAL1-E	"				
Fuse	100	3.49	4.86	21.10 ± 0.50	590. ± 12.
87CAL2-F	**Green amphiboles**	(total fusions)			
Fuse	100	49.62	4.64	53.00 ± 5.89	1224. ± 99.
87CAL2-H	"				
Fuse	100	27.45	14.33	25.22 ± 2.88	685. ± 65.
87CAL2-I	"				
Fuse	100	30.44	9.70	29.36 ± 1.20	776. ± 26.
87CAL2-J	"				
Fuse	100	11.28	4.99	22.60 ± 3.04	625. ± 71.
87CAL2-K	"				
Fuse	100	26.33	3.79	25.32 ± 5.52	687. ± 124.

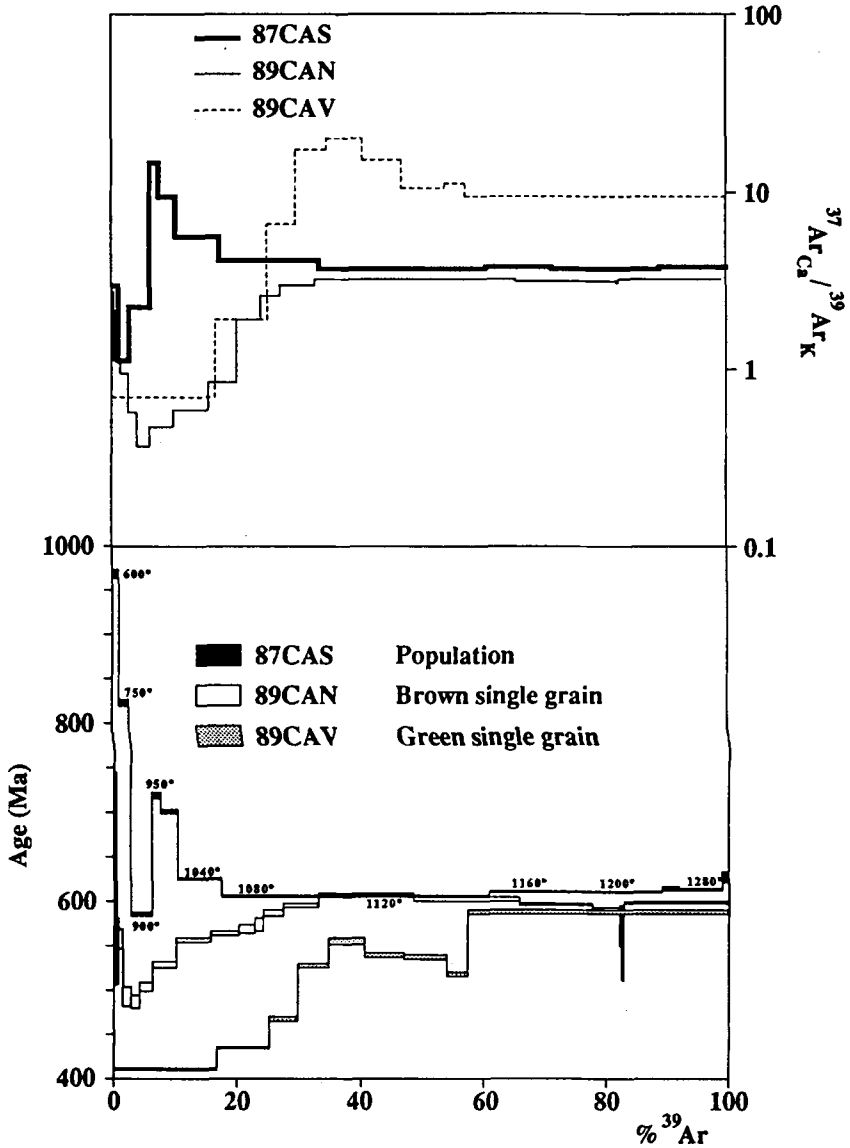


Figure 2. Age and $^{39}Ar_K/^{37}Ar_{Ca}$ spectra for samples 87CAS, 87CAN and 87CAV. Brackets for apparent ages represent 1σ errors. Temperatures for experiment 87CAS are given in °C.

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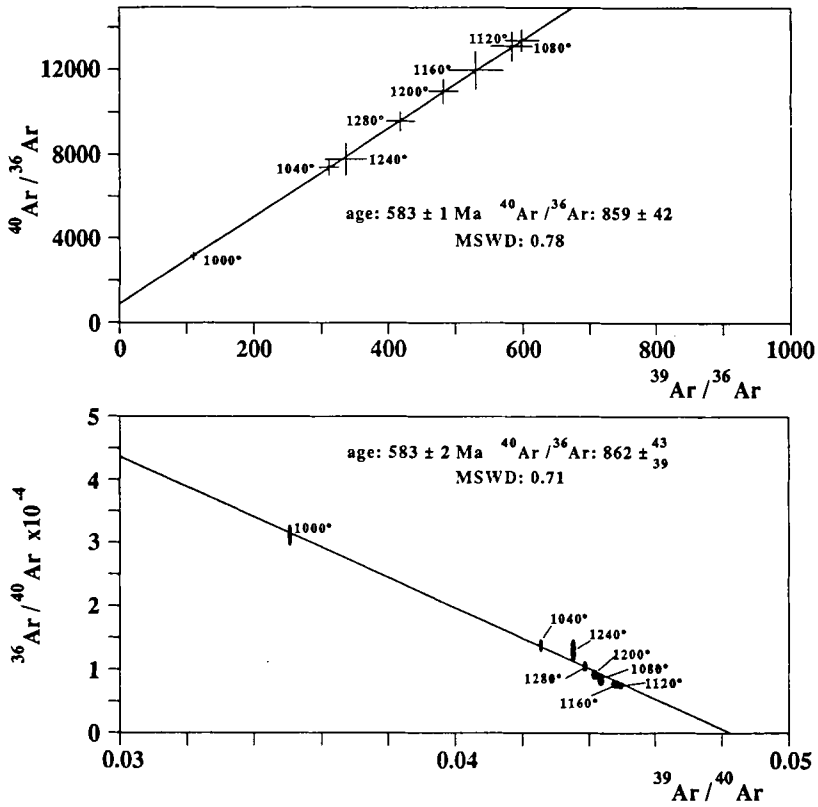


Figure 3. $^{40}Ar/^{36}Ar$ versus $^{39}Ar/^{36}Ar$ isochron and $^{36}Ar/^{40}Ar$ versus $^{39}Ar/^{40}Ar$ correlation diagrams for sample 87CAS.

Laser step heating of a fragment of green amphibole (89CAV; see Fig. 2)

We obtained a very irregular age spectrum that yields a total integrated age of 526.0 ± 0.7 Ma. The first 40 per cent of $^{39}Ar_K$ released (five steps) show an increase of the ages from 411 to 555 Ma that corresponds to an increase of the measured $^{37}Ar_{Ca}/^{39}Ar_K$ ratios from low values (Ca/K = 1.2), probably due to secondary K-rich products, to high values (Ca/K = 35.4) characteristic of Ca-rich amphiboles. In the remaining 60 per cent of $^{39}Ar_K$ released, the $^{37}Ar_{Ca}/_{39}Ar_K$ ratios decrease regularly (down to Ca/K = 16.9) whereas the ages, always lower than 590 Ma, display an irregular behaviour.

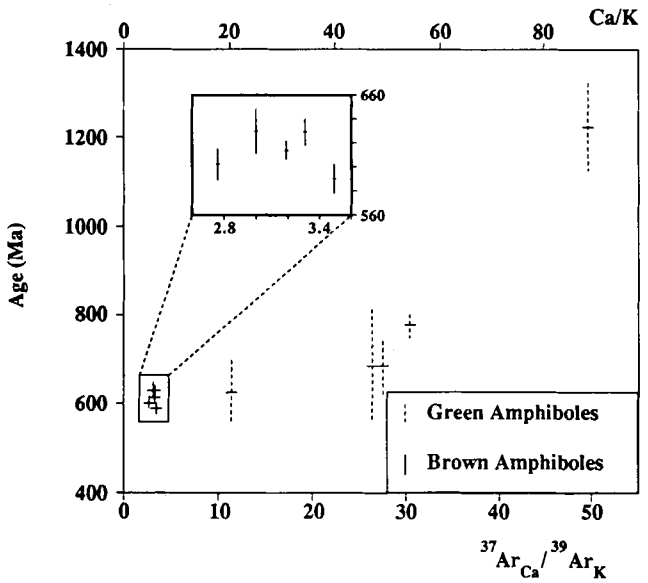


Figure 4. Laser fusion experiments on pieces of amphibole. The ages obtained are plotted versus the measured Ca/K ratio (Ca/K = 1.79 × $^{37}Ar_{Ca}/^{39}Ar_K$).

Interpretation of the $^{40}Ar/^{39}Ar$ dating results

The presence of excess argon in the amphiboles of the Keralain diorite is deduced from (1) the saddle-shaped age spectrum displayed by the step heating experiment performed on the population of amphiboles (87CAS) and (2) the corresponding high initial $^{40}Ar/^{36}Ar$ ratio deduced from the correlation and isochron calculations.

In spite of the flatness of the saddle-shaped age spectrum which should be characteristic of a low amount of excess argon, there is a significant difference (22 Ma) between the isochron age (which may represent the true age) and the minimum age from the age spectrum 87CAS. This behaviour may be due to similar degassing patterns of both radiogenic and excess ^{40}Ar during the step-heating experiment. This phenomenon has been observed on amphiboles by Onstott, Sisson & Turner (1989).

The laser experiments were performed to locate this excess argon in the heterogeneous population of amphiboles. The excess argon can be distributed:

(i) in the whole crystals of hornblende or in the green rims only,

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(ii) in some specific minerals only, or
(iii) in the whole population of amphiboles.

The combined use of laser and electron microprobe analysis allows the following interpretations.

(1) The single-step fusion ages obtained on the pieces of brown amphibole (core of the hornblende) (weighted mean: 609 ± 5 Ma) and the saddle shape of the age spectrum displayed by the single grain of brown amphibole (89CAN) argue in favour of the presence of argon in excess in the core of the hornblendes. Nevertheless, this excess is relatively lower than in the population (87CAS) since the minimum age (590.8 ± 2.3 Ma) displayed by the single-grain age spectrum (89CAN) is (a) lower than that (604.9 ± 1.2 Ma) of the population (87CAS) and (b) concordant at the 2σ level with the calculated correlation age (583 ± 2 Ma) obtained from the population.

(2) The microprobe analysis performed on the green rims of the hornblende (Ca/K ratios of the order of 11) show that the green fragments of amphibole analysed by laser single-step fusion (with higher Ca/K ratios) do not correspond to the rims of the hornblende but may be fragments of actinolite or actinolitic hornblende. The high anomalous total fusion ages are due to a higher concentration of excess argon and/or a lower amount of potassium (and then radiogenic ^{40}Ar) than in the hornblende. Nevertheless, the step-heating experiment performed on a fragment of green amphibole (89CAV) showed (a) no evidence of excess argon, (b) an integrated age of 526.0 ± 0.7 Ma, much younger than that of the other green grains and (3) a total integrated $^{37}Ar_{Ca}/^{39}Ar_K$ ratio of 8.7, lower than that of the other fragments of green amphiboles. This different behaviour may be due to a higher degree of alteration to potassic secondary products. On the other hand, the age of the fusion step (588.0 ± 1.4 Ma: 42.6 per cent of the total $^{39}Ar_K$ released) is concordant at the 2σ level with the calculated correlation age.

These observations show that the whole population of amphiboles is affected, at various degrees, by the presence of argon in excess. On the other hand, we have a concordance of the ages (at 2σ level) between (a) the correlation (and isochron) age of 583 ± 2 Ma obtained from the population, (b) the minimum age of 590.8 ± 2.3 Ma of the flat saddle-shaped age spectrum displayed by the fragment of brown hornblende, and (c) the fusion step age of 588.0 ± 1.4 Ma obtained on a fragment of green amphibole. Nevertheless, we suggest that the best estimate of the age of the amphiboles of the Keralain diorite is given by the calculated correlation age (583 ± 2 Ma).

Isotopic closure temperature

The isotopic closure or blocking temperature T_c has been calculated using the iterative formula (Dodson 1973):

$$T_c = \frac{E}{\left| R \ln\left(A\tau \frac{D_0}{a^2}\right)\right|}$$

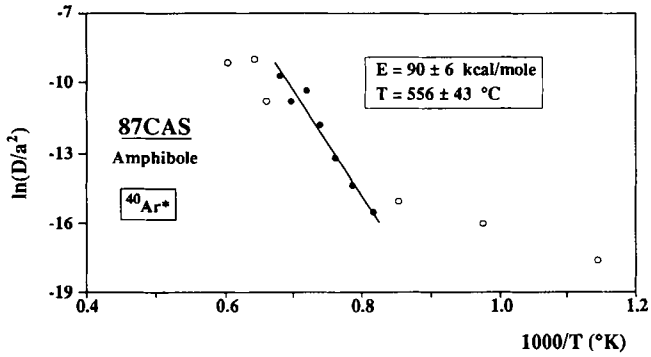


Figure 5. Arrhenius plot of calculated D/a^2 values from the gas released data of sample 87CAS versus absolute and reciprocal absolute temperature. See text for explanations.

where

$$\tau = \frac{-RT_c^2}{\left(E\frac{dT}{dt}\right)}$$

is the cooling time constant, E the activation energy, dT/dt is the cooling rate, R is the gas constant, D_0/a^2 is the pre-exponential frequency factor and 'a' a characterisitic dimension of the mineral grain (see Fig. 5).

The parameters D/a^2 were calculated from the relative volumes of ^{40}Ar released between 950 °C and 1200 °C (81.5 per cent of the $^{39}Ar_K$ released) using the formulae compiled by Musset (1969). A volumic diffusion with a cylindrical geometry was assumed for the hornblende (Berger & York 1981b).

The activation energy and the pre-exponential factor have been calculated from an Arrhenius plot by least-squares fitting (York 1969) of the diffusion coefficients to a straight line segment. The calculated activation energy (E = 90 ± 6 kcal mol^{-1}) is on the same order as the results presented by Berger & York (1981b), Lopez-Martinez & York (1982) and Freer (1981). The calculations performed with the relative volumes of $^{39}Ar_K$ released give similar results and therefore show that the excess argon does not seem to significantly affect the calculation of the diffusion parameters. The calculated isotopic closure temperature (T_c = 556 ± 43 °C) is also in close agreement with values obtained or presented by the above authors. This temperature must be considered as an estimate because in these calculations, it is assumed that (1) the mineral remains stable over the considered temperature interval and (2) the Ar is released by volumic diffusion, assumptions which are difficult to verify. On the other hand, Harrison (1983) mentioned the capacity of amphibole to readjust to the vacuum environment, imposed by the experimental procedure, without significantly altering its internal Ar isotope distribution.

PALAEOMAGNETISM

Laboratory procedures

Natural remanent magnetization measurements were made with a digital Schonstedt DSM-1 spinner magnetometer. Alternating field demagnetization was done with a

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Schonstedt GSD-1 and thermal demagnetization was performed in a non-magnetic furnace (Schonstedt TSD-1) with cooling in zero field. The residual field during experiments and measurements was always lower than 10 nT.

Natural remanent magnetization

A short-period (8 days) viscosity test (Thellier & Thellier 1959) has been applied to 64 specimens cut from the 31 cores drilled at Keralain. The intensities of the natural remanent magnetization (NRM) have a mean value of 5×10^{-2} A m^{-1}. The viscosity coefficients have a moderate mean value of 16 per cent. NRM directions are very well grouped and the mean direction obtained from the 64 specimens is $D = 305.6°$, $I = 46.6°$, $k = 44$, $\alpha_{95} = 2.7°$.

At Petit Taurel, the same viscosity test was performed for eight specimens. We obtained viscosity coefficients around 35–40 per cent and a larger scatter in the NRM directions, with a mean: $D = 263.3°$, $I = 76.5°$, $\alpha_{95} = 27.9°$ $k = 4.9$.

Demagnetizations

Keralain

Demagnetization at high alternating field (AF) values was not possible for the Keralain diorite samples because of the appearance of a parasitic component of magnetization (Fig. 6a). Moreover at the maximum peak field value of 100 mT, a large part of the NRM remains. Nevertheless AF demagnetization up to 30 mT was efficient in removing a steep downwards component that corresponds to the present-day field direction. Due to this behaviour thermal demagnetization was applied, to one specimen from each sample from Keralain.

A soft component is generally removed by 270 °C for specimens with a low viscosity coefficient (< 20 per cent) (Fig. 6c) and by 330 °C for specimens with higher viscosity coefficient (> 20 per cent). This component is generally steep and downwards and probably corresponds to the present-day field. It is easily removed by AF demagnetization (Fig. 6a) and is therefore interpreted as a viscous remanent magnetization (VRM).

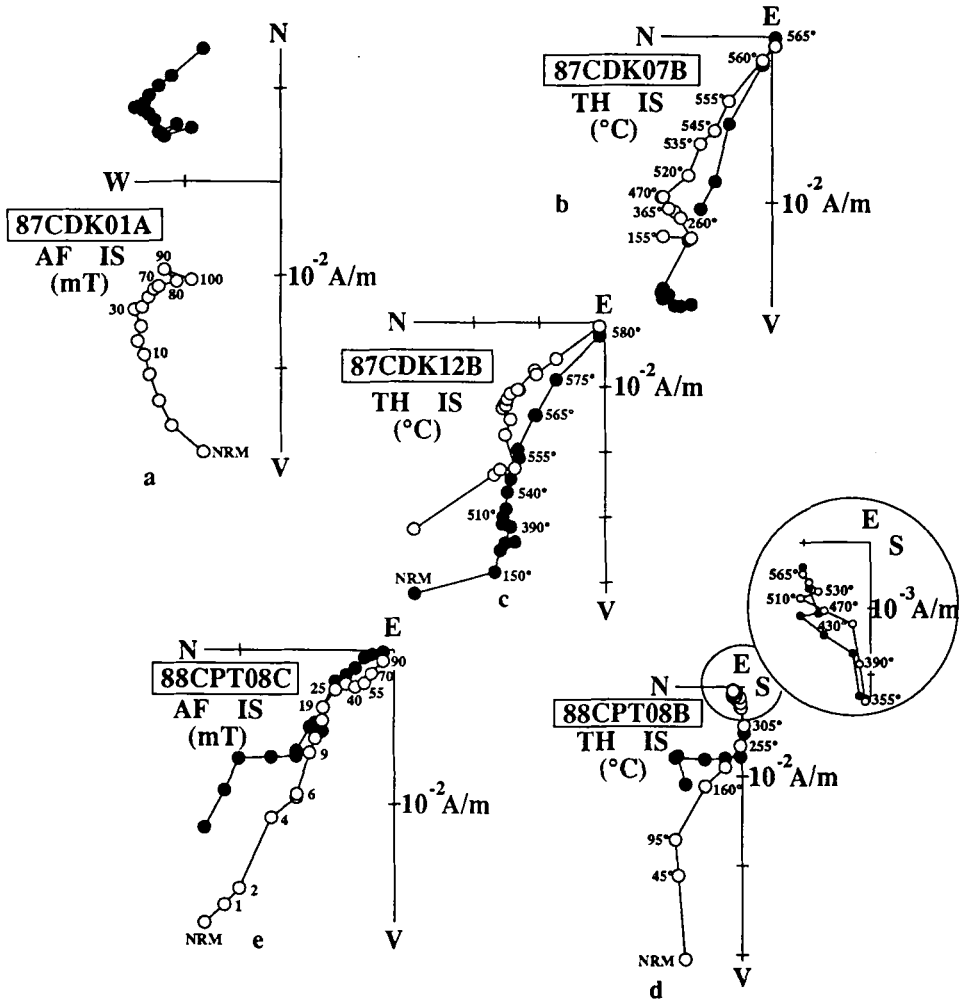


Figure 6. Zijderveld demagnetization diagrams. Closed and open symbols represent respectively the projections of vector endpoints on the horizontal and vertical planes. AF and TH indicate alternating field or thermal treatment. Blow-up is shown for (d).

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Between 300° and 580 °C, one (Fig. 6b) or two components (Fig. 6c) of magnetization can be isolated. Generally, a stable endpoint is reached above 500 °C and then the magnetization component decays towards the origin, indicating that the high-temperature component is well isolated.

Petit Taurel

As for Keralain one specimen from each sample was subjected to thermal demagnetization. Indeed, AF demagnetization reveals a complex behaviour (Fig. 6e) compared to thermal demagnetization (Fig. 6d), probably due to overlapping coercivity spectra.

Two, or more often three, components (Fig. 6d) can be isolated from thermal demagnetization. A soft component, steep and downwards, is generally removed by 50° or 100 °C and has probably a laboratory viscous origin. The intermediate component, removed by 300° or 350 °C, corresponds to the *in situ* present-day field, whereas the final and characteristic component has unblocking temperature up to 580 °C. For one specimen, an increase of the magnetic susceptibility appears after 440 °C, together with erratic behaviour. Generally, the demagnetization trajectories do not quite go to the origin in the Zijderveld diagrams (Fig. 6d).

Rock magnetism

Thermomagnetic experiments under vacuum conditions were performed to analyse the carriers of the magnetization.

Keralain

A typical magnetite Curie temperature around 580 °C was observed (Fig. 7a). Regardless of a relatively important paramagnetism, there is no clear evidence for other carriers of magnetization apart from magnetite. The hardness of the magnetization with respect to AF demagnetization suggests that magnetite is present in the form of single-domain grains.

Reflected light examination of polished sections has revealed non-titaniferous magnetite grains (≈50 μm), together with fresh pyrite and a few goethite grains produced by alteration. Some grains of chalcopyrite and rutile (in the inclusions of the hornblende) were also observed.

In summary, we consider that the magnetic carrier of the characteristic high-temperature magnetization is fine-grained magnetite.

Petit Taurel

As at Keralain, the thermomagnetic experiment (Fig. 7b) defines a Curie temperature around 580 °C indicating that magnetite is one of the magnetic carriers. Nevertheless the shape of the thermomagnetic curve between 100° and 400 °C suggests the presence of a second magnetic carrier (maghaemite?). This might explain the peculiar behaviour of specimens during AF demagnetization (Fig. 6e).

Polished section examinations show that large mineralogical transformation phenomena have affected the whole rock. Pyrite, chalcopyrite and a few goethite grains were observed, but the only magnetic carrier is magnetite sometimes slightly martitized.

Isothermal remanent magnetization (IRM) acquisitions, performed on samples from Keralain (87CDK21A) and the Petit Taurel (88CPT01A), reveal saturation fields (Fig. 8) typical of magnetite (200–300 mT) and thus confirm the thermomagnetic experiments.

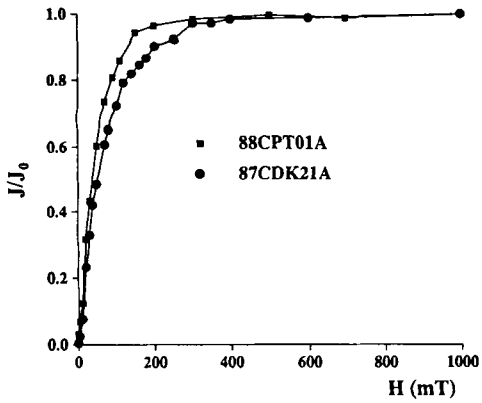


Figure 8. Normalized intensity of Isothermal Remanent Magnetization versus applied field.

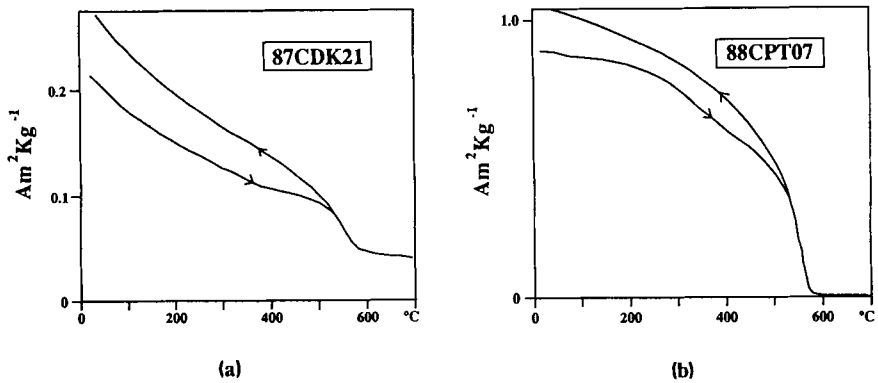


Figure 7. Results of thermomagnetic experiments using a Curie balance.

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Table 2. Site-mean characteristic directions. N_S is the number of samples per site used to compute the mean and n/n_0 is the ratio of specimen directions used to compute the mean to the total number of analysed specimens. Declination and inclination values are in degrees. k and a_{95} are statistical parameters associated with the mean (Fisher 1953). VGP gives the virtual geomagnetic pole position computed from the mean direction.

Site	Site coordinates	N_S	n/n_0	Magnetic Direction (D/I)	k	a_{95}	VGP
Keralain	48.79°N,3.09°W	31	31/31	300.4/33.5	63.5	3.2	33.6°N,256.2°E
Petit Taurel	48.77°N,2.95°W	6	6/9	219.7/34.1	43.5	10.3	13.8°S,318.5°E

DISCUSSION OF THE RESULTS

The demagnetization trajectories were analysed from Zijderveld diagrams and the characteristic components of magnetization were determined graphically by line fitting. A total of 38 out of 40 samples have given demagnetization diagrams for which one or two characteristic components can be reliably isolated (see Table 2)

The 31 samples from Keralain exhibit a coherent and well-grouped high-temperature magnetization. These directions have been plotted in Fig. 9(a). The mean direction, labelled A, calculated from the 31 Keralain samples is $D = 300.4°$, $I = 33.5°$, $\alpha_{95} = 3.2°$. The corresponding palaeopole (DKA) is located at 33.6°N, 256.2°E. It closely corresponds to the GK_1 component previously determined at Keralain by Hagstrum *et al.* (1979) (Fig. 9b). However, the mean directions and circles of confidence of these two studies are slightly but significantly different. We shall discuss the origin of this discrepancy later, together with the significances of the secondary components.

The characteristic component from the Petit Taurel appears in six samples with unblocking temperatures up to 580 °C. The mean direction named B is fairly well defined with $D = 219.7°$, $I = 34.1°$, $\alpha_{95} = 10.3°$ (Fig. 9b).

Because of the 'overshooting' in the demagnetization trajectories and the mineralogical observations, we suspect that the B component of magnetization is a quasi-total remagnetization. Moreover, a high-temperature component (isolated close to the Curie temperature of magnetite) has been observed in only one sample [88CPT08B, Fig. 6(d), $D = 329.5°$, $I = 26.9°$. It is similar to the component A isolated from Keralain samples. The component B defines a palaeopole (DKB) located at 13.8°S, 318.5°E.

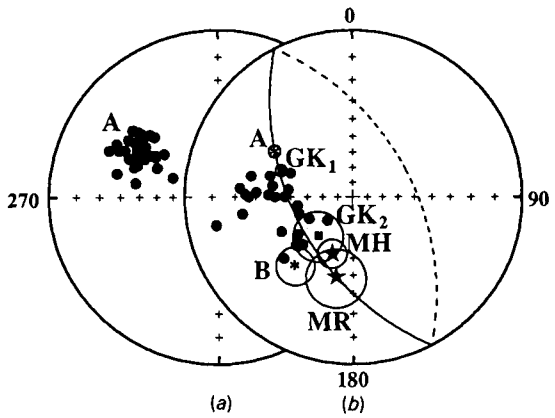


Figure 9. Stereographic projection. (a) High-temperature directions from Keralain. (b) Mean characteristic directions (A and B) from this study (asterisks), secondary directions from Keralain (dots) and mean directions (GK_1 and GK_2) from previous study (squares) (Hagstrum *et al.* 1979). MR and MH represent the mean directions from comparative studies (see text for explanations)

This remagnetization can be related to the appearance of an alkali granite-ignimbritic activity in the Domnonean domain at the end of the Cadomian history, the different forms of which have been dated at 547 ± 56 Ma and 552 ± 8 Ma respectively the 'rhyolites ignimbritiques de Lézardrieux' and the 'microgranites de loguivy' :Rb–Sr(WR); Auvray 1975; Vidal 1980). Previous palaeomagnetic studies of this magmatism have defined palaeopoles for the 'rhyolites ignimbritiques de Lézardrieux' (Hagstrum *et al.* 1979) and for the 'microgranites de Loguivy' (Perigo *et al.* 1983), named respectively MR and MH. The corresponding directions of magnetization are plotted in Fig. 9(b). The component B seems quite close to the MR direction. Its high unblocking temperature (up to 560–580 °C) implies, if it is a thermoremanent remagnetization, that the stock of the Petit Taurel has been heated at least up to 530 °C for 1 Ma or 550 °C for 1 yr (Pullaiah *et al.* 1975). Such high temperatures would certainly have provoked important mineralogical transformations, such as those observed in the present samples under the microscope.

A component similar to B appears in 24 samples from Keralain with intermediate unblocking temperatures (440–540 °C). The directions, plotted in Fig. 9(b), do not present a Fisherian distribution but are rather scattered along the great circle joining the directions A (Keralain high temperature) and MR. This indicates that thermal demagnetization has not been effective in isolating this partial remagnetization from component A, and suggests a chemical or thermochemical origin rather than purely thermal. None of these directions is more southward than direction B; there is therefore no reason to suspect that B has not been correctly isolated. Hagstrum *et al.* (1979) have observed a similar secondary component (GK_2). Its mean direction has been plotted in Fig. 9, and lies on the same great circle trend.

From this study and previous work, all the observed components, namely A, B, GK_1, GK_2, and secondary components defined above, are situated along a great circle including also the directions from the younger Cadomian intrusive bodies (MR and MH). It is therefore highly probable that all intermediate directions result from a mixture of two components, (1) the primary component of the Keralain diorite and (2) a thermochemical remagnetization induced by the late intrusions of the alkaline magmatism around 550 Ma. In this respect, it appears that the difference between A and GK_1 could correspond to a better isolation of components in the present study.

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All these observations confirm the primary origin of the direction A isolated in all the samples from the stock of Keralain. The magnetic carrier is known to be magnetite and the blocking temperature is estimated at 575 °C. The isotopic closure temperature of 556 ± 43 °C calculated from the $^{40}Ar/^{39}Ar$ step-heating experiment performed on amphibole is not significantly different from the magnetic blocking temperature. Furthermore, on account of the small size of the stock, the decrease in temperature must have been rapid. Thus the magnetic and isotopic systems were probably closed at the same time, so that an age of 583 Ma can be assigned to the A component of magnetization.

IMPLICATION FROM THIS STUDY

See Fig. 10 and Table 3. The Domnonean domain comprises several structural units between which motions have occurred principally during the Palaeozoic. Nevertheless these movements, small in magnitude, are assumed to be negligible for our purposes as suggested by Hagstrum *et al.* (1979).

From the compilation of the palaeomagnetic results for the Armorican Massif (Perroud 1985), a set of Late Precambrian palaeopoles can be identified, whose age is poorly constrained. The palaeopole DKA, defined from the present study, falls within this cluster. As it is precisely dated, it gives the opportunity to calibrate the dating of the Armorican polar wander path.

The DKA palaeopole lies close to the *in situ* (PP_1) or dip-corrected (PP_2) poles from the St Peter Port Gabbro of Guernsey (Hailwood & Garrett 1977), the age of which is between 660 and 570 Ma (Adams 1976). It is also very close to the palaeopole SQ_2 (Hagstrum *et al.* 1979), defined by the secondary magnetization from the diorite de St Quay Portrieux. However, its age of 584 ± 56 Ma (Vidal *et al.* 1972) (Rb–Sr: WR) is assumed to be the age of the primary magnetization that defined a palaeopole SQ_1 located close to the palaeopoles SP (Spillites de Paimpol) (Hagstrum *et al.* 1979) and KMB (Kératophyres de tréguier) (Perigo *et al.* 1983) whose age is assumed to be 642 ± 46 Ma (Rb–Sr: WR) (Vidal 1980). This suggests that the diorite de St Quay Portrieux could be older that the above-mentioned age of 584 Ma, while being within the relatively important 2σ errors interval.

In this case, the SQ_2 pole may correspond to a remagnetization related to the appearance of an important calc-alkaline magmatism (at about 585 Ma) during the Cadomian orogeny (Guerrot & Peucat 1990), corresponding with emplacement of the Keralain diorite. It seems probable that the secondary magnetization of the Kératophyres de Tréguier (palaeopole KHI) (Perigo *et al.* 1983) is due to this same event.

The palaeopole JD (Duff 1980), defined from the Jersey group C dolerites also falls within this group. However, its Cadomian age seems unlikely because these dykes intersect the SE granite whose age is 515 ± 26 Ma or 500 ± 11 Ma (Adams 1967) (Rb–Sr: respectively WR and Bi + WR).

While the effect of the Cadomian orogeny are best preserved in the Armorican Massif, they are also observed in the southern part of the British Isles. As these two domains have very similar geological histories, it seems important to estimate their relative positions.

The DKA palaeopole seems in rather good agreement with palaeopoles of similar age from southern England and Wales (south of the Iapetus suture). In fact, the MO_1 palaeopole from the Gwna Group pillow lavas of Anglesey (Piper 1976), for which an age of 580–600 Ma is assumed (Moorbath & Shackleton 1966) (K–Ar and Rb–Sr), is very similar.

Furthermore, Lomax & Briden (1977) and Piper (1982) have both performed palaeomagnetic studies of the Malvernian complex (Welsh Borderlands, Wales) for which they defined respectively palaeopoles MA_1 (tilt corrected Malvernian palaeopole 13°N, 252°E) and M_2 (29.5°N, 275.7°E). A geochronological study by Lambert & Rex (1966) (K–Ar and Rb–Sr) concluded that a major isotopic event occurred at around 600 Ma. Fitch *et al.* (1969)

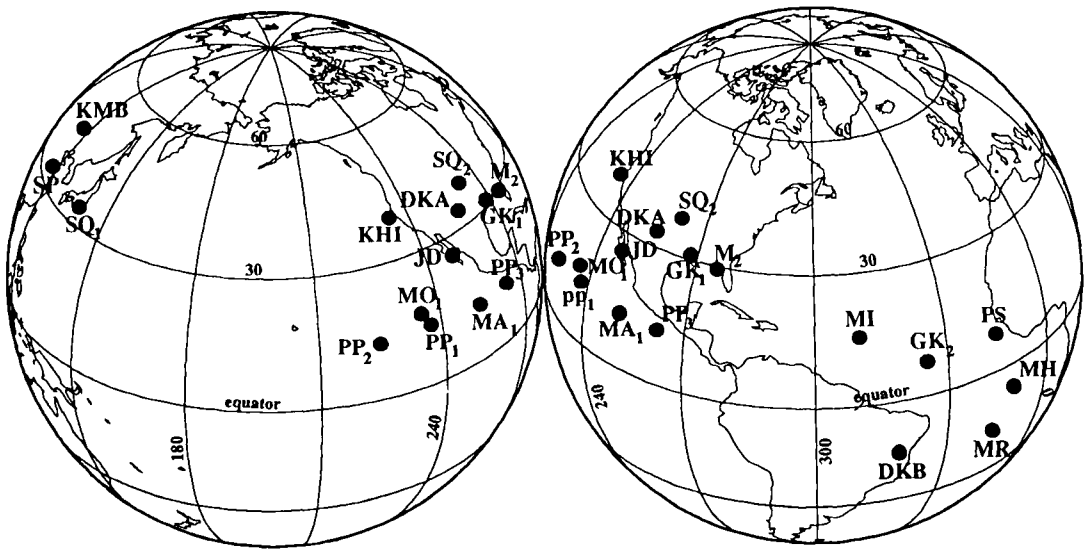


Figure 10. Palaeomagnetic pole positions as listed in Table 2, plotted on stereographic projections. Note that the two diagrams partly overlap.

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Table 3. Late Precambrian palaeomagnetic pole positions from the Armorican Massif, southern England and Wales.

SYMBOL	UNIT	AGE (MA)	POLE POSITION	REFERENCE
ARMORICAN MASSIF, FRANCE				
SP	**Spilites, Paimpol**	642 ± 46	34°N,117°E	Hagstrum et al.(1980)
	Keratophyres, Tréguier	642 ± 46		Perigo et al.(1983)
KMB	Magnetite		45°N,116°E	
KHI	Hematite		39°N,234°E	
	Diorite, St Quay	584 ± 56		Hagstrum et al.(1980)
SQ_1	-High blocking T°		34°N,134°E	
SQ_2	-Intermediate blocking T°			
	Gabbro, St Peter Port	570-660		Hailwood and Garrett(1977)
PP_1	-In Situ		14°N,237°E	
PP_2	-Bedding corrected		13°N,224°E	
PP_3	-Bordeaux diorite contact		13°N,263°E	
	Diorite, Keralain			
DKA	-High blocking T°, Keralain	583 ± 2	34°N,256°E	This study
DKB	-Remagnetization, Petit Taurel		14°S,319°E	This study
GK_1	-High blocking T°, Keralain		31°N,268°E	Hagstrum et al.(1980)
GK_2	-Intermediate blocking T°, Keralain		8°N,325°E	Hagstrum et al.(1980)
JD	**Dykes dolerites, Jersey**	PK-K	26°N,248°E	Duff (1980)
PS	**Granite, Porz-scarff**	552 ± 8	9°N,343°E	Hagstrum et al.(1980)
MR	**Mainland rhyolites**	547 ± 56	16°S,343°E	Hagstrum et al.(1980)
	Microgranite, Loguivy			
MH	-Dykes microgranite		6°S,340°E	Perigo et al.(1983)
MI	-Contact dykes microgranite		16°N,310°E	Hagstrum et al.(1980)
SOUTHERN ENGLAND and WALES				
MO_1	**Mona Complex, Gwna**	580-600	17°N,235°E	Piper (1976)
MA_1	**Malvernian Complex**	≈ 600	13°N,252°E	Lomax and Briden (1977)
M_2	**Malvernian Complex**	≈ 600	30°N,276°E	Piper (1982)

reported similar K–Ar mineral ages for the Malvernian Complex. A Rb–Sr study has been carried out by Beckinsale *et al.* (1981). They define an isochron age of 681 ± 53 Ma from samples of the Malvernian igneous complex that range from ultramafic diorite to granite. It seems unlikely that these rocks have a cogenetic origin so that the 'isochron' defined is probably an 'errorchron'. This implies that the age of the palaeopoles MA_1 and M_2 is not 681 ± 53 Ma as suggested by Lambert & Rex (1966).

These two palaeopoles are in close agreement with the palaeopoles of similar age from the Armorican Massif (PP_1 or PP_2, SQ_2, DKA, KHI). The palaeolatitudes calculated from these palaeopoles and from the palaeopoles MO_1, MA_1 and M_2 give respectively 15°S and 9°S. These similarities could indicate that the southern United Kingdom and the Armorican Massif were coupled together around 600 Myr ago.

These observations reinforce the assumption of Hagstrum *et al.* (1979) that the southern United Kingdom, Czechoslovakia and the Armorican Massif behaved as a coherent continental mass, the 'Armorica plate', during Latest Precambrian and Early Palaeozoic times. The Cadomian orogeny, from 640 to 540 Ma, is characterized by a large and rapid apparent polar wander.

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